

RED LAKE, ON, Canada

WMO: 718540

Lat: 51.070N

Lon: 93.790W

Elev: 1266

StdP: 14.04

Time Zone: -6.00 (NAC)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.5	-24.0	-35.8	0.8	-28.2	-31.3	1.0	-23.5	19.9	6.1	17.8	7.0	4.2	250	0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	82.8	65.9	79.8	64.7	76.9	62.7	69.5	77.9	67.3	75.0	65.4	73.2	8.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.7	103.0	73.4	64.7	96.1	70.7	62.7	89.4	69.0	34.4	78.3	32.6	74.7	31.1	73.0	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.7	16.6	14.6	-34.5	88.2	3.4	3.3	-37.0	90.6	-39.0	92.5	-40.9	94.4	-43.4	96.8		
(5)			-34.7	73.4	3.4	3.3	-37.2	75.8	-39.1	77.8	-41.0	79.7	-43.5	82.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.3	0.6	3.3	19.2	33.6	49.9	60.1	64.2	62.8	51.6	38.1	20.9	5.5		
	DBStd	25.27	14.49	13.16	13.30	11.39	10.02	6.80	5.75	6.31	8.58	8.40	11.78	14.89		
	HDD50	7147	1532	1307	956	502	130	6	0	1	80	380	874	1379		
	HDD65	11355	1997	1727	1421	943	478	175	86	118	409	833	1324	1844		
	CDD50	1423	0	0	0	9	125	309	440	399	129	12	0	0		
	CDD65	156	0	0	0	0	9	28	60	51	8	0	0	0		
	CDH74	1345	0	0	0	0	138	269	466	427	45	0	0	0		
	CDH80	249	0	0	0	0	26	43	88	87	5	0	0	0		
	WSAvg	7.3	6.9	7.0	7.5	7.6	7.3	7.1	6.9	6.5	7.8	8.1	8.1	6.8		
	PrecAvg	26.3	1.3	0.9	1.3	1.6	2.2	3.8	3.5	3.6	3.0	2.1	1.9	1.3		
(15) Precipitation	PrecMax	33.7	3.2	2.8	3.2	4.6	4.4	6.9	6.5	6.4	5.8	7.5	4.1	3.0		
	PrecMin	15.5	0.2	0.2	0.2	0.2	0.4	1.3	1.4	0.9	0.3	0.3	0.5	0.5		
	PrecStd	4.1	0.7	0.5	0.7	1.2	1.1	1.4	1.3	1.4	1.3	1.5	1.1	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	42.1	50.2	68.8	83.6	84.6	86.7	86.3	79.8	66.2	55.1	37.3		
		MCWB	34.1	35.6	40.8	53.4	62.0	64.2	68.6	66.9	66.7	55.8	48.8	33.3		
	2%	DB	32.4	35.9	44.6	61.2	77.4	80.0	82.4	82.3	73.0	60.4	44.1	31.4		
		MCWB	31.3	32.0	36.4	48.7	59.0	62.7	66.8	66.8	63.5	52.7	40.7	29.9		
	5%	DB	26.6	31.9	40.6	55.8	73.3	76.9	78.9	78.9	68.7	55.8	40.3	28.9		
		MCWB	25.4	29.2	34.8	43.9	56.8	61.5	64.7	65.5	59.7	49.7	38.0	28.0		
	10%	DB	21.4	24.4	37.1	51.6	67.9	73.8	76.1	75.2	65.3	51.3	35.3	26.3		
		MCWB	20.2	22.1	32.6	41.7	54.7	59.6	63.2	64.1	58.8	45.8	33.1	25.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.7	37.2	41.5	55.5	66.8	68.8	72.5	71.8	69.3	57.9	48.2	33.3		
		MCDB	35.8	41.3	47.9	66.1	75.2	77.0	81.2	79.8	76.5	64.5	54.5	36.9		
	2%	WB	31.6	32.5	37.9	49.5	63.1	65.7	69.6	69.4	65.3	53.8	42.0	30.2		
		MCDB	32.5	35.3	42.9	57.7	72.0	75.1	77.9	78.0	69.8	58.5	44.0	31.3		
	5%	WB	25.5	28.6	35.4	45.6	59.3	63.8	67.3	67.2	62.3	50.1	38.0	28.1		
		MCDB	26.7	32.0	39.4	54.8	70.6	72.3	74.6	75.2	67.0	54.8	40.0	28.9		
	10%	WB	20.1	22.4	33.1	41.9	55.8	61.9	65.3	65.3	58.9	46.0	33.5	25.4		
		MCDB	21.3	24.3	36.7	50.8	66.9	70.0	72.8	73.2	63.8	51.4	35.2	26.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.3	20.6	21.9	21.1	21.7	20.5	18.8	19.5	16.3	12.7	11.9	14.2		
		MCDBR	16.9	21.2	20.6	27.4	28.3	25.8	23.5	23.9	21.7	20.8	14.1	11.2		
		MCWBR	16.2	18.3	14.4	16.4	13.7	11.9	11.7	11.7	13.0	13.3	10.9	9.7		
	5% WB	MCDBR	16.9	20.5	17.5	24.9	23.4	20.2	19.3	20.8	18.2	19.0	13.1	10.2		
		MCWBR	16.2	17.8	12.5	16.2	12.8	11.1	11.3	12.1	12.5	13.0	10.6	8.9		
(40) Clear-Sky Solar Irradiance	taub		0.248	0.266	0.286	0.323	0.342	0.362	0.386	0.359	0.336	0.293	0.275	0.240		
	taud		2.293	2.261	2.332	2.419	2.467	2.435	2.382	2.473	2.528	2.602	2.444	2.338		
	Ebn at Noon		252	276	292	292	290	284	274	277	272	269	241	238		
	Edh at Noon		20	28	32	33	33	35	36	31	26	20	18	16		
(44) All-Sky Solar Radiation	RadAvg		358	733	1138	1607	1682	1723	1766	1472	1046	596	314	241		
	RadStd		42	60	92	114	109	146	143	127	117	73	33	31		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page